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Description

 [Equivalent Materials to Stainless steel UNS S31603 as a PDF file](#)

Type 420F stainless steel is chromium, general-purpose steel, which can be heat treated. Due to the addition of sulfur in its composition, it provides greatly improved machinability.

Annealed, it offers good ductility. It renders excellent resistance to corrosion in the hardened condition.

It is similar to type 420, but stainless-steel type 420F is preferred in applications where good machining properties are required due to complex geometries, such as dental equipment, surgical equipment.

Common Names:

Stainless steel type 420F is commonly known as Chromium steel.

Executive Standards/Grades:

- ASTM: A895
- UNS: S42020
- DIN: 1.4028
- AISI: 420F
- AMS: 5620

Chemical Properties:

Type 420F stainless steel has the following chemical composition:

- Carbon – 0.15%
- Manganese – 1.00%
- Chromium – 13%
- Sulphur – 0.15%
- Phosphorus – 0.040%
- Silicon – 1.00%
- Molybdenum – 0.60%
- Iron – 84%

Mechanical Properties:

- Hardness (HRC) – 18
- Elongation (%) – 22
- Yield Strength (Mpa) – 380
- Elastic Modulus (Gpa) – 210
- Tensile Strength (Mpa) – 655
- Poisson's Ratio – 0.27-0.30

Physical Properties:

- Density (x1000 kg/cm³) – 7.8
- Thermal Conductivity (BTU in/hr.ft². °F) – 174
- Specific Heat (Btu/lb/°F – [32-212 °F]) – 0.11
- Electrical Resistivity (microhm-cm, at 68 oF) – 330
- Thermal expansion (µin/in°F) – 5.72
- Melting Point – 2650 °F

Key Features:

- Excellent machinability properties, due to added sulfur.
- Similar to type 420, but with free machining properties
- Provides excellent resistance to corrosion when hardened
- Can withstand heat treatment
- Annealed, it provides good ductility
- Not usually welded, due to hardening properties
- Only minor cold work operations can be performed because it risks cracking

Forms Available: Stainless steel type 420F is available in different forms such as:

- Pipe
- Plate
- Bar
- Wire
- Tube
- Flanges
- Fittings

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Applications:

Because of its properties, stainless steel type 420F has a wide range of applications including:

- Cutlery
- Dental and surgical equipment
- Pump shafts
- Gears
- Hand tools
- Steel balls
- Cams
- Pinions

Possible Alternatives Grades:

Type 420F stainless steel has some possible alternatives that can be used. Such as:

- 420 – Less machinable.
- 420J2 – Excellent corrosion, low cost.
- 430F – Good machinability
- 440F – High carbon, maximum hardness

FAQs:

- ***Can stainless steel type 420F be hot worked and cold worked?***

Type 420F stainless steel can withstand minimal cold work because it risked cracking. For hot working, it is recommended slowly increase the temperature to 1400°F, then to 2000-2200°F. To prevent cracking, let it cool slowly in the furnace.

- **Can type 420F be welded?**

Type 420F stainless steel is not usually welded, due to its hardening characteristics to air. However, if welding is necessary, it should be preheated to 300-400°F and then tempered for 2 hours.

- **How resistant is type 420F to corrosion?**

Type 420F stainless steel provides excellent resistance to corrosion when it is in the hardened condition.

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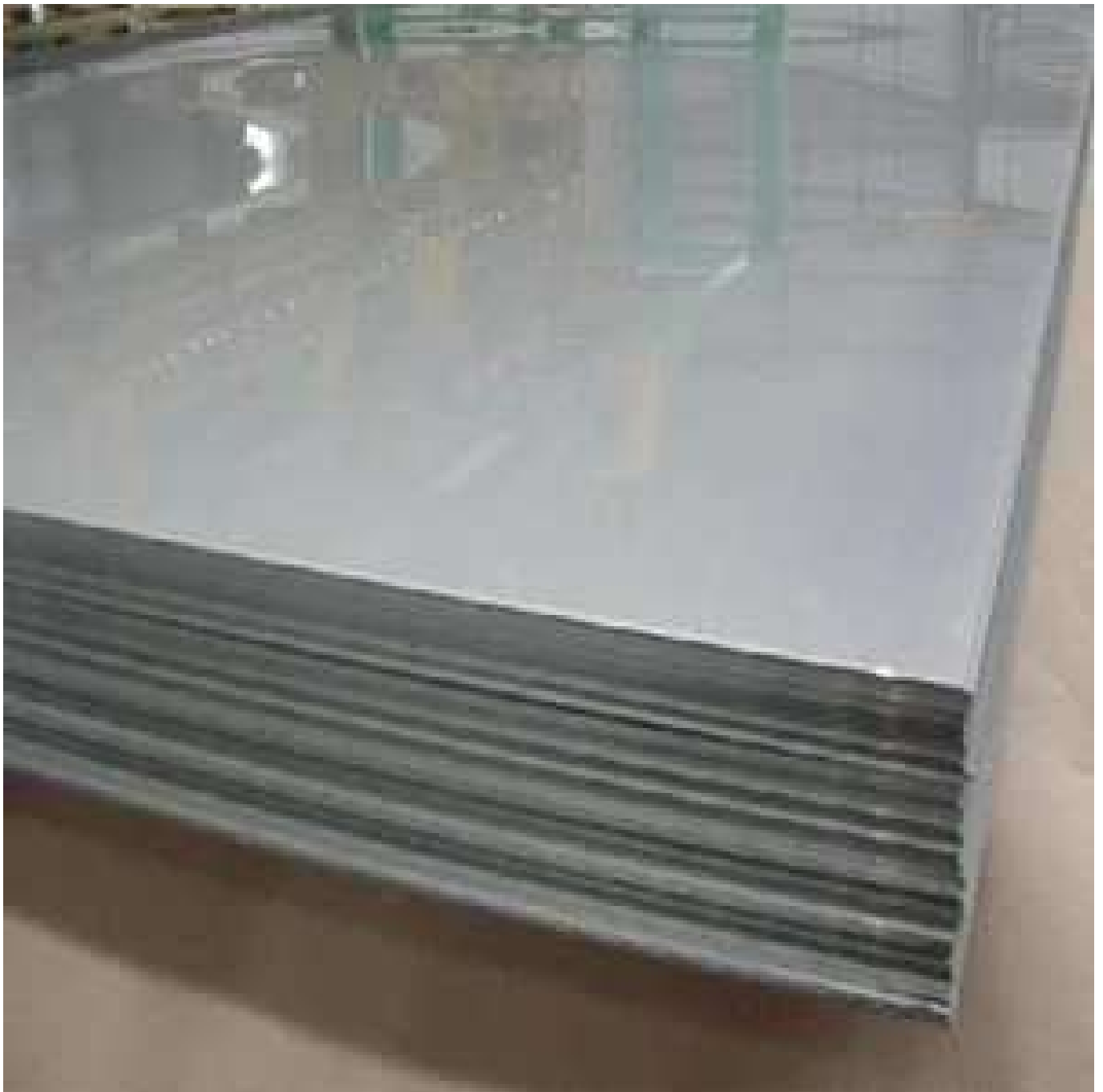
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